Lecture 0 - Outline (modified)

ECE 447: Data Analysis and Machine Learning for Engineers

Bahareh Tolooshams, Ph.D.
Assistant Professor,
University of Alberta
btolooshams@ualberta.ca
https://btolooshams.github.io

January 06, 2026

Land Acknowledgment

The University of Alberta acknowledges that we are located on Treaty 6 territory, and respects the histories, languages, and cultures of First Nations, Métis, Inuit, and all First Peoples of Canada, whose presence continues to enrich our vibrant community.

In-Context Learning Capabilities of Large Language Models (LLM)

Prompt

France	→	Paris
Japan	→	Tokyo
China	→	Beijing
Canada	→	...

Completion Ottawa

How can we explain this astonishing capabilities?

One explanation: LLM are based on ***transformers****, and transformers can construct* ***linear regression*** *based on the context data, and do* ***gradient descent*** *on that (to* ***optimize*** *it)!*

Transformers Learn In-Context by Gradient Descent

Johannes Von Oswald, Eyvind Niklasson, Ettore Randazzo, Joao Sacramento, Alexander Mordvintsev, Andrey Zhmoginov, Max Vladymyrov *Proceedings of the 40th International Conference on Machine Learning*, PMLR 202:35151-35174, 2023.

Why *Should We Learn Foundations of*
Data Analysis *and* ***Machine Learning***
instead of a Deep Learning or LLM Course?

Course Setup

Welcome to ECE 447: Data Analysis and Machine Learning for Engineers!

Administrative information:

- Instructor: Bahareh Tolooshams (office hours: Wednesday 9:15 am - 10:30 am)
- Teaching assistant (TA): Jaeho Chung
- Our primary communication channel is Canvas.

Prerequisites:

- ECE 220 or CMPUT 275 —> Software/coding skills for data analysis

and

- ECE 342 or STAT 235 —> Introduction to probability for machine learning

Meet Your Instructor ...

I am the PI of **NeuBahar** Lab.

Neuro–Bayesian AI for Human-interpretable Abstractions and Representation Learning

https://btolooshams.github.io/bahar.html

We are interested in artificial and biological (brain) Intelligence:

Representation learning

Interpretability

Generative models

Neuroscience

Logistics

Lectures

- Location: CSC B-10
- Date: Tuesdays and Thursdays
- Time: 11:00 AM - 12:20 PM

Seminars

- Location: CCIS 1-160 (starting tomorrow)
- Date: Wednesdays
- Time: 08:00 AM - 08:50 PM

Assessment

• Assignments (4)	20%
• Project	25%
• Midterm	25%
• Exam	30%

Read the Outline and Syllabus (on Canvas) Carefully!

Class Participation / In-class Quiz

Each lecture and seminar will contain in-class question(s).

This is **ungraded** and **voluntary**.

Lectures and seminars will ***both*** include voluntary ungraded in-class quizzes!

Assignments

Mixture of

- Pen & paper (math behind machine learning methods).
- Data analysis.
- Software (coding) implementation of machine learning methods.

Course Project

Groups of size 3.

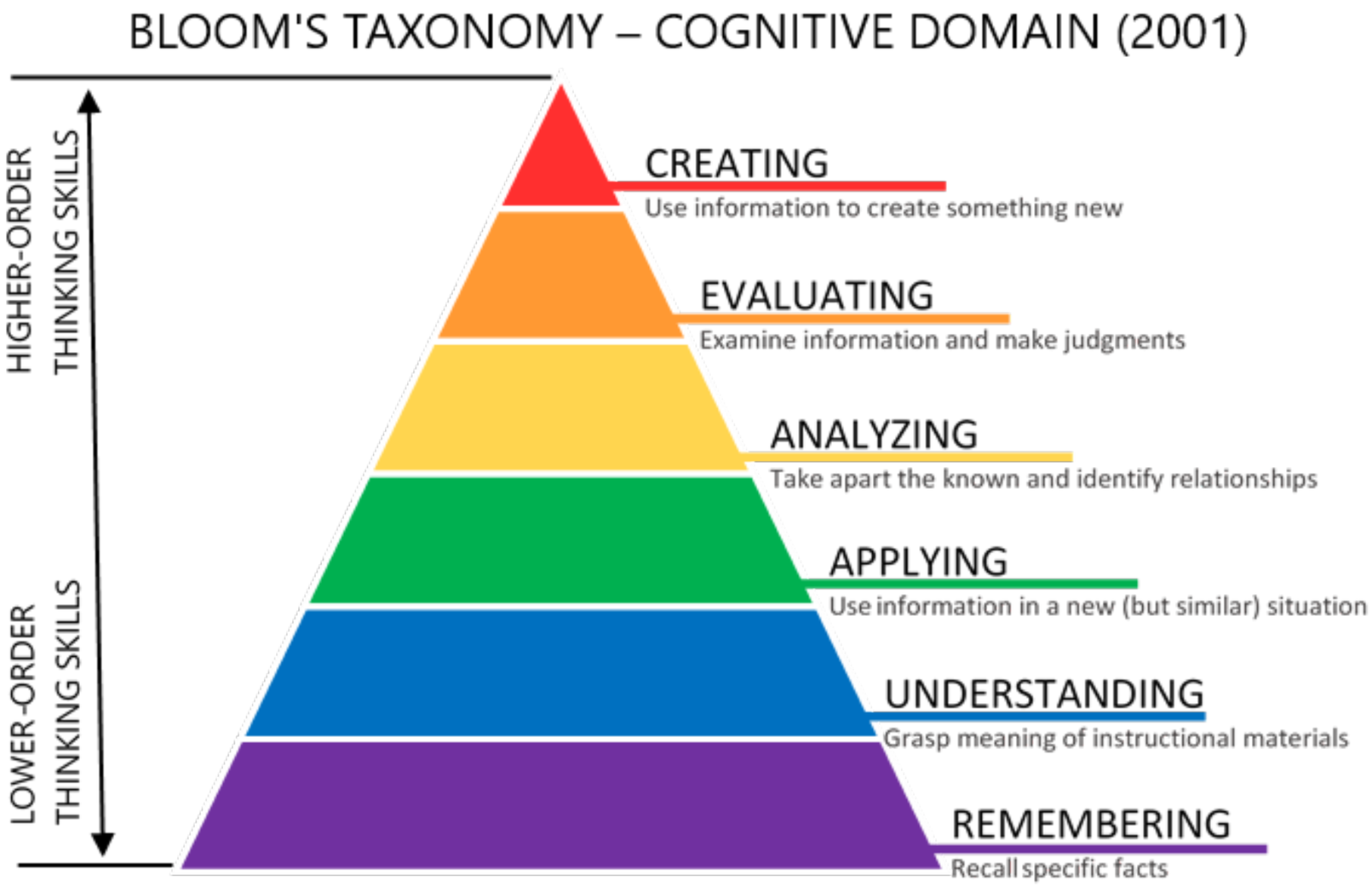
You will **present and reproduce** a published machine learning paper.

- In-class presentation (hence, your classmates will learn about your topics).
- Final report.

There will be a seminar info session about course project.

Aim for **deeper** understanding of the concepts.

Do not memorize!



Source: https://citt.ufl.edu/resources/the-learning-process/designing-the-learning-experience/blooms-taxonomy/blooms-taxonomy-graphic-description/

Course Objectives

Goal I

To teach you basics of data analysis

Goal II

To expose you to algorithms and techniques of Machine Learning (ML)

Goal III

To teach you to mathematical concepts helpful in working with data analysis/processing methods

Goal IV

To give you the background required for using ML Operationalization Management (MLOps)

Lecture Topic Guide

Read the Outline and Syllabus for more information!

Part I: Machine Learning

- Data analysis and evaluation
  Predictive analytics, data exploration, data quality, data processing, training/testing, performance measures, model comparison
- Linear models
  Linear regression, linear discriminant functions, minimum (weighted) distance classifier
- Information-based learning
  Decision trees, entropy, information gain, tree pruning, random forest, ...
- An optimization model for classification
  Support vector machines, feature space, polynomial regression, kernels
- Probabilistic machine learning
  Linear regression, cost function, gradient descent, logistic regression, regularized optimization
- Unsupervised learning
  K-means algorithms, hierarchical clustering, dimensionality reduction
- Neural networks
  Perceptron, neural network, backpropagation and gradient descent
- Emerging topics in deep learning
  Sparse autoencoders and interpretability

Lecture Topic Guide

Part II: Machine Learning Life Cycle

- Introduction to Machine Learning Operationalization Management (MLOps)

  Machine learning pipelines, model evaluation and packaging, deployment of ML system, testing and securing ML solution, model serving and monitoring, continual learning

Seminar Topic Guide

Part III: Some Math

- Linear algebra
- Analytic geometry
- Probabilities
- Distributions
- Matrix decomposition
- Vector calculus
- Optimization

Part III: Some Programming

- Basic of Python
- Numerical libraries
- Visualization libraries

Programming Environment

- Jupyter Notebook
- Google Colaboratory
- CLion (Python IDE)

Student's question:

"I thought we are going to do programming, but the seminars/lectures seem to cover lots of math"

Instructor's reply:

"In this course, we will work with data and also do programming (coding) in Python. However, ***prior to that*** *you will gain insights about each machine learning methods using tools from linear algebra, probability, and statistics."*

Flow of Lectures/Seminars

1. Mathematical concepts required (primarily covered in Seminars).

2. Mathematical formulations behind machine learning (during Lectures).

3. Coding examples on how the methods work (during Lectures/Seminars).

My Recommendation to My Students for Research

1. Formulate problems on paper.
2. Understand and analyze the formulation.
3. Make sure to deeply know what the method is supposed to do.
4. Transfer the formulation into a programming language such as Python.
5. Verify that it performs as expected.

Welcome to ECE 447!

Questions?

Course Planning Form

https://forms.gle/Cm7Yj9DNGT8Gb3XG7